Introduction to ROSE5770
Unmanned Systems

Instructor:

Prof. Chen Xi (Patrick), SHB601E, Tel: 39430814, xichen002@cuhk.edu.hk

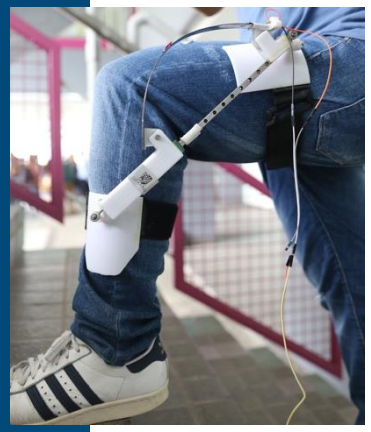
Tutor:

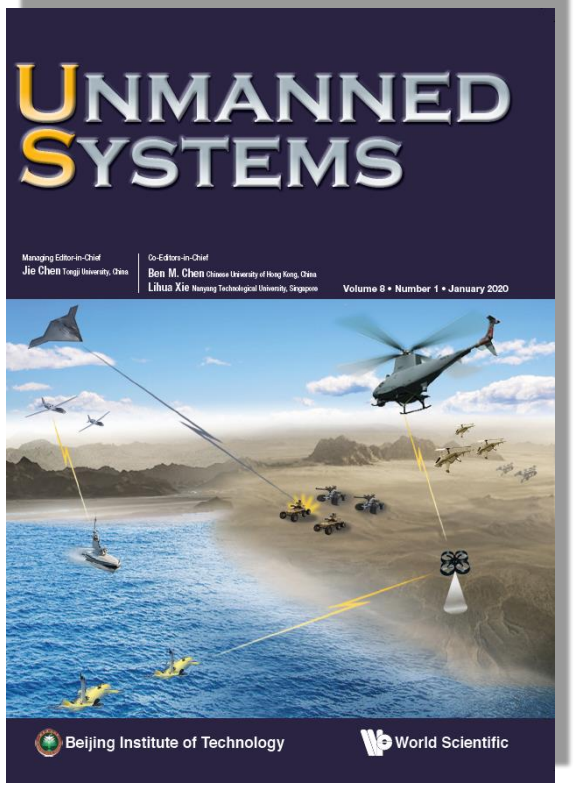
Zhipeng Guo, zhguo2@mae.cuhk.edu.hk

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Course Introduction







"Unmanned Systems" is a cutting-edge course as a response to the rapid technological advancements and increasing industry demands in this domain. This course aims to provide students with a comprehensive understanding of theoretical foundations and practical implementations of unmanned systems, for example unmanned aerial vehicle (UAV), unmanned ground vehicles (UGV), unmanned surface and underwater vehicles (USV, UUV)…

The course is specifically designed to expose students to the challenges and opportunities present in the design, control, and application of unmanned systems, thereby preparing them for impactful careers in this rapidly evolving field.

The course will cover essential technical elements including introduction to unmanned systems, **system architecture, autonomous control, sensors for unmanned systems, sensor fusion, path planning, navigation systems, AI for unmanned systems**, and the future challenges and opportunities.

Expected Learning Outcomes

1) Understand the essential components and design principles of unmanned systems, including their types, characteristics, and applications.
2) Gain knowledge of autonomous control algorithms and levels, as well as system stability factors in unmanned systems.
3) Acquire skills in integrating and using various types of sensors in unmanned systems and learn data fusion techniques for sensor data.
4) Master various path planning algorithms and obstacle avoidance techniques in dynamic environments.
5) Develop an understanding of the role and application of AI in unmanned systems, including machine learning algorithms and AI decision-making processes.



- ✓ Highlight real-world applications.
- ✓ How unmanned systems are designed, integrated and deployed to solve real-world challenges in smart cities.

Course Assessment

- **Individual Assessment** (30%)
  - ✓ Short-essay
  - ✓ Late submission: subject to 80% penalty (permission required)
- **Quiz (20%)**
  - ✓ 1 in-class test (multiple choice)
- **Mini-project (50%)**
  - ✓ Group of **4-5** members with 1 group leader (**Group leaders** email group info (list of name, student ID) on or before **Sep 25, 2026)**
  - ✓ Oral Presentation (20%)
  - ✓ Written Report (30%)
- **10 Bonus Credits (based on in-class/out-of-class performances)**

❖ The ***final group mini-project*** is related to the unmanned system applications. Every student needs to ***orally present*** their project results. The group report needs to clearly indicate the ***contribution of each member***.

❖ Always clearly identify your ***Names***, ***Student No.*** OR ***Group Number*** and ***Member Names*** on the cover of any submission.

Academic Honesty

- Students are expected to conform to the highest standards of honesty and integrity
- Teamwork is strongly encouraged in this class, but the final HW/report must reflect their originality and individual efforts
- For further details, please refer to
  - Honesty in Academic Work: A Guide for Students and Teachers (http://www.cuhk.edu.hk/policy/academichonesty)
  - Short videos to understand Academic Honesty/Plagiarism (http://www.cuhk.edu.hk/clear/tnl/acad_honesty.html)
  - VeriGuide: Plagiarism Detection (https://academic2.veriguide.org/portalcuhk/)

- **Use of Generative AI tools:**
  - Use of AI tools is not allowed.

Course Schedule

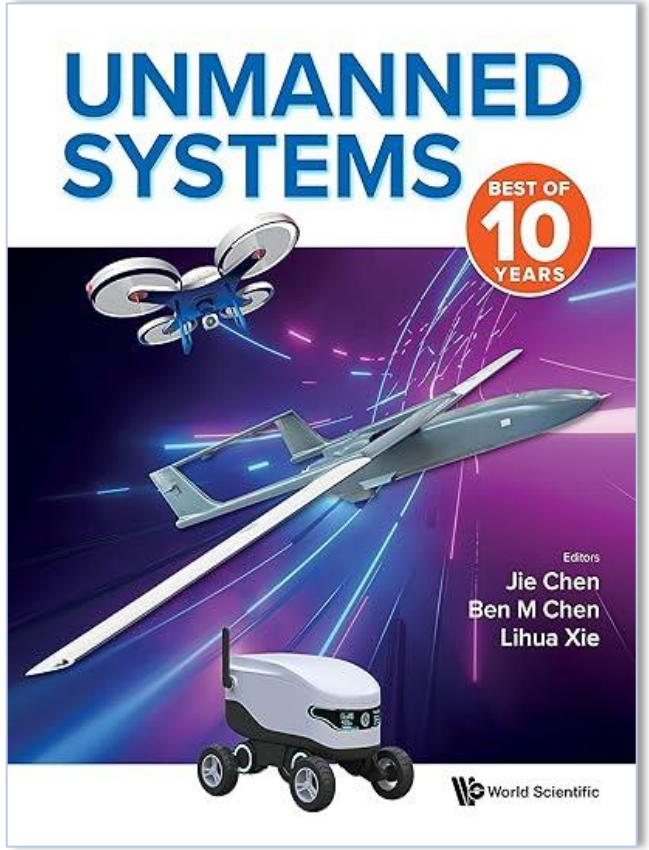
Week	Date	Topic	Assessment
1	Sep 7	Introduction to Unmanned Systems	
2	Sep 14	Introduction to Unmanned Aerial Vehicles (UAVs)	
3	Sep 21	UAV: system architecture (to be switched with Nov 2)	UAV application example
4	Sep 28	UAV: autonomous navigation	
5	Oct 5	UAV: sensor system	Individual Assignment (tentative)
6	Oct 12	UAV: SLAM (Simultaneous Localization And Mapping)	
7	Oct 26	UAV applications and future trends I	
8	Nov 2	UAV applications and future trends II	
9	Nov 9	AI for UAVs	Quiz (tentative)
10	Nov 16	Unmanned Surface Vehicles	
11	Nov 23	Underwater Vehicles	
12	Nov 30	Autonomous Driving Vehicles	
13	Dec 7	Project presentation I	Group project
14	Dec14	Project presentation II	Group project

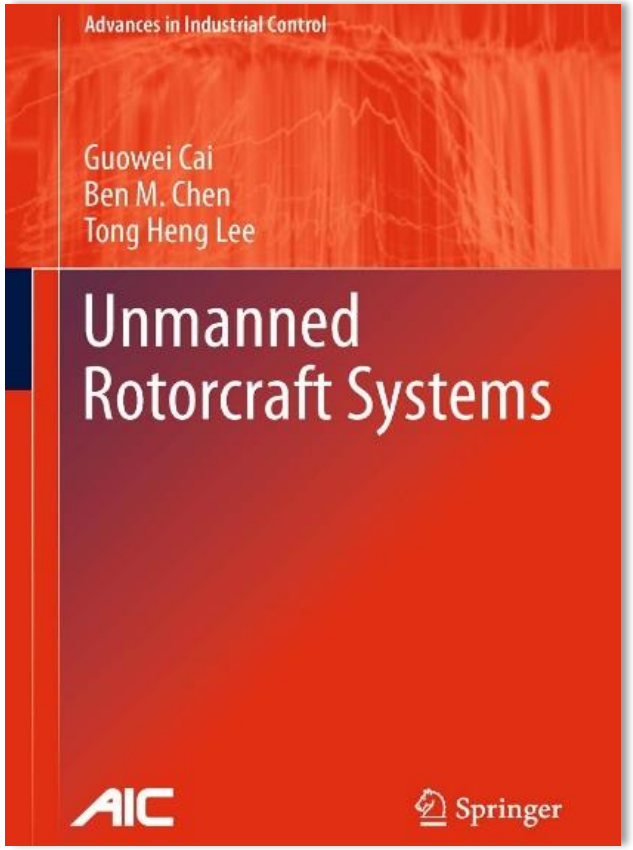
References (No set books)

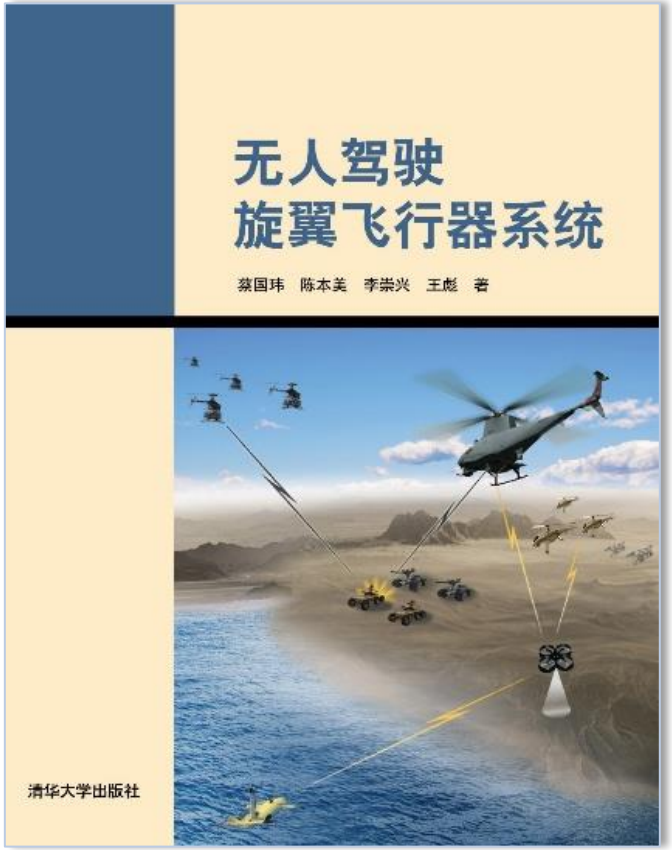
[1] Unmanned Systems, An international journal published by WSPC & BIT Press.

[2] Guowei Cai, Ben M. Chen, Tong Heng Lee. Unmanned Rotorcraft Systems, Springer, 2011.

[3] 蔡国玮，陈本美，李崇兴，王彪. 无人驾驶旋翼飞行系统，清华大学出版社，2012



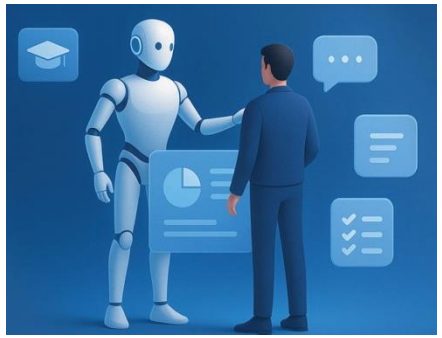




Overall framework of unmanned systems

1. Hardware and software systems
2. Dynamic modeling
3. Flight control systems
4. Perception and localization
5. Motion planning
6. Task/mission planning
7. Real-world applications

AI







Basic characteristics

Core Features

Autonomy
Ability to operate independently without human intervention

Sensing Capabilities
Various sensors for environment perception and data collection

Decision-Making
Processing information to make operational decisions

Mobility
Ability to move freely within operational constraints

Communication
Data exchange with other systems and human operators

Technical Challenges

Environmental Perception
Accurately interpreting complex and changing environments

Real-time Decision Making
Processing data and making decisions under time constraints

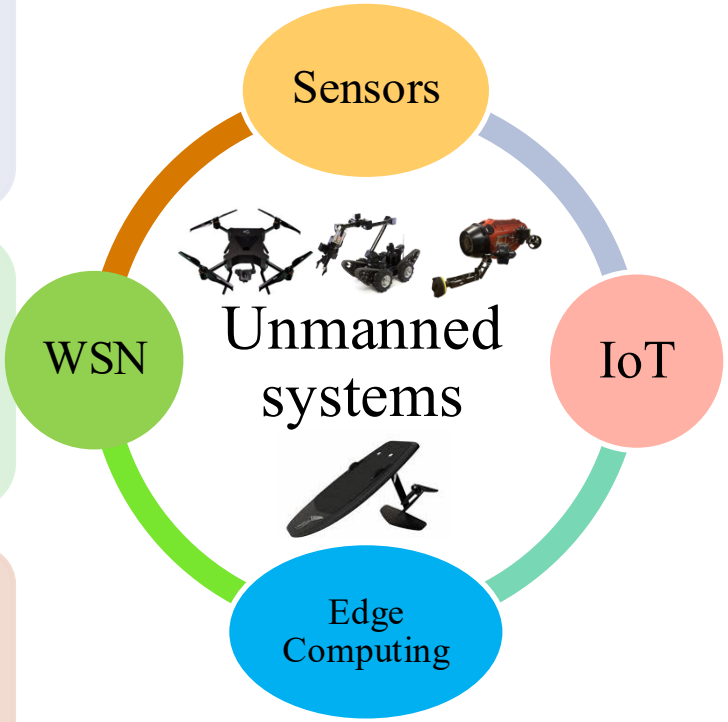
System Reliability and Safety
Ensuring consistent operation in critical situations

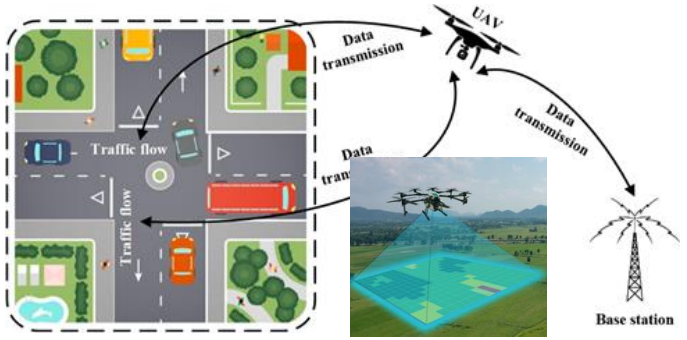
Energy Management
Optimizing power consumption for extended operations

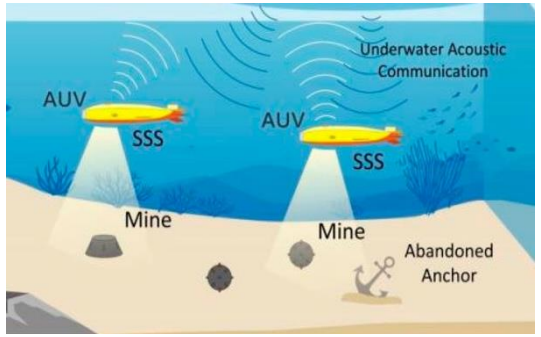
Human-Machine Interaction
Effective collaboration between humans and autonomous systems

Applications

Data collection Inspection Monitoring	• Environmental survey & monitoring • Traffic monitoring • Civil infrastructure inspection & monitoring • Post-disaster survey
Interactive Operation	• Public safety (law enforcement, emergency response, search & rescue, firefighting) • Logistics and transportation (medical, food, passenger) • Agriculture • Military and defense • Urban environment management (maintenance)
Others	• STEM education • Competition & Recreation • Academic research • Journalism











Classifications



UGV



UAV







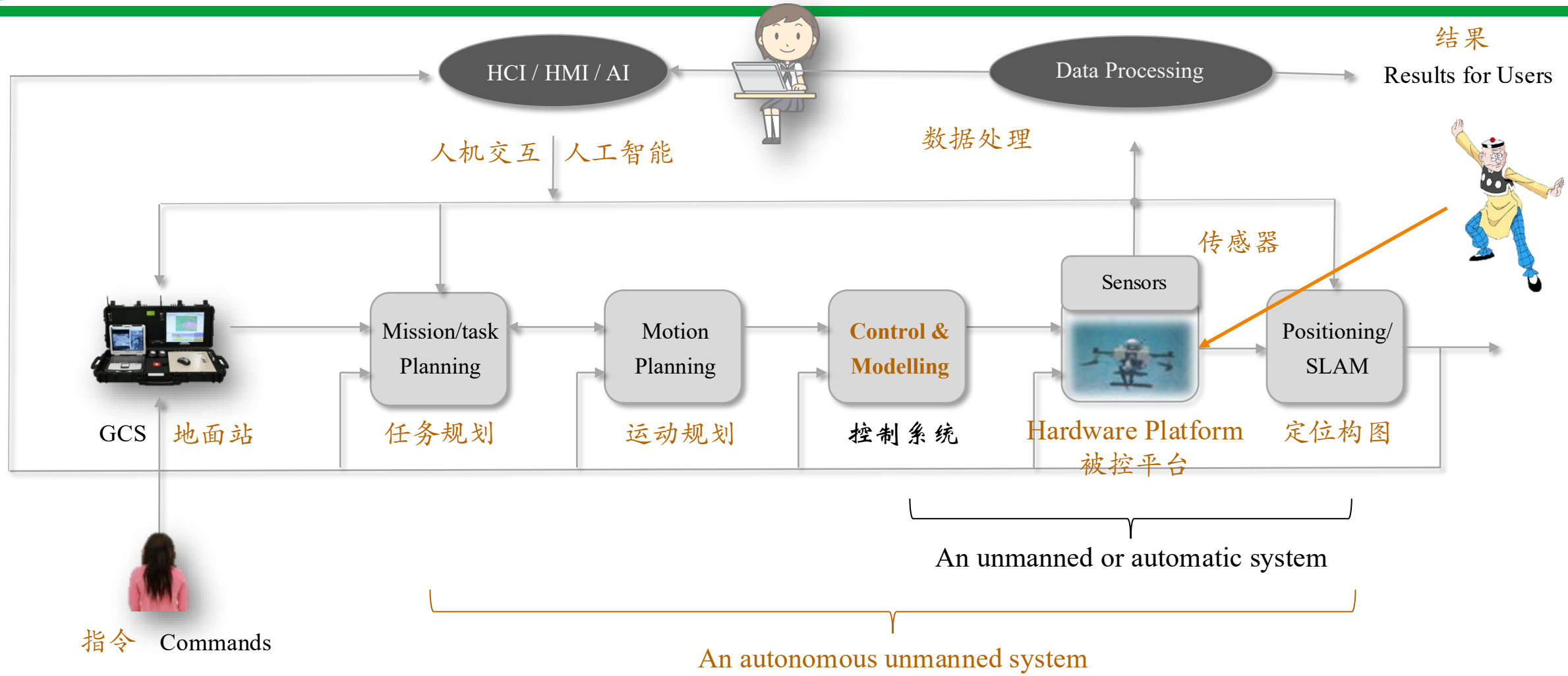
USV



UUV



Intelligent autonomous unmanned system framework



Humans (the intelligent agents) are usually the most difficult to control!

Hardware platform construction



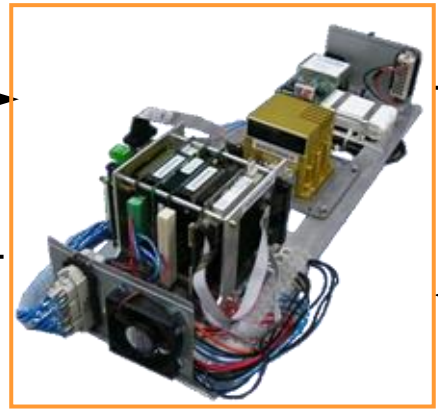
The 1st UAV at NUS 2003

NUS National University of Singapore

It is also one of not too many in the world in early 2000…

Command

Real-time Data



Onboard System

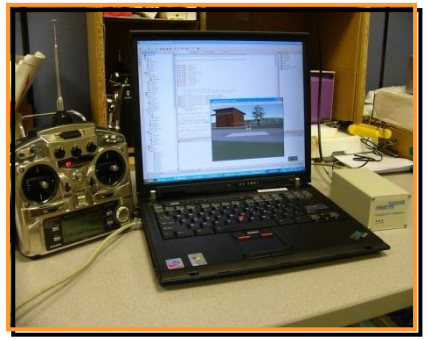
Control Signal

Measurement

Bare Helicopter



Manual Operation



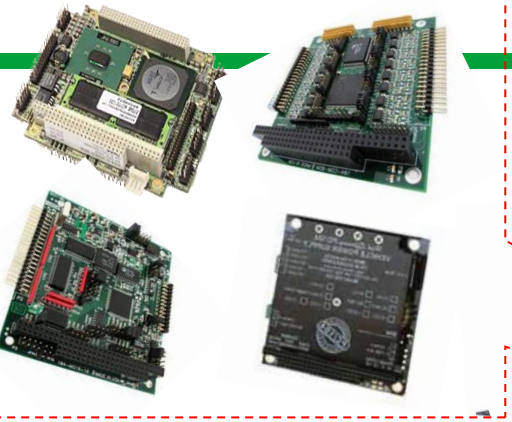
Ground Station

we built ⟸ = = ⟹ we bought



RC Joystick

Hardware platform construction



PC-104 Computer Stack:

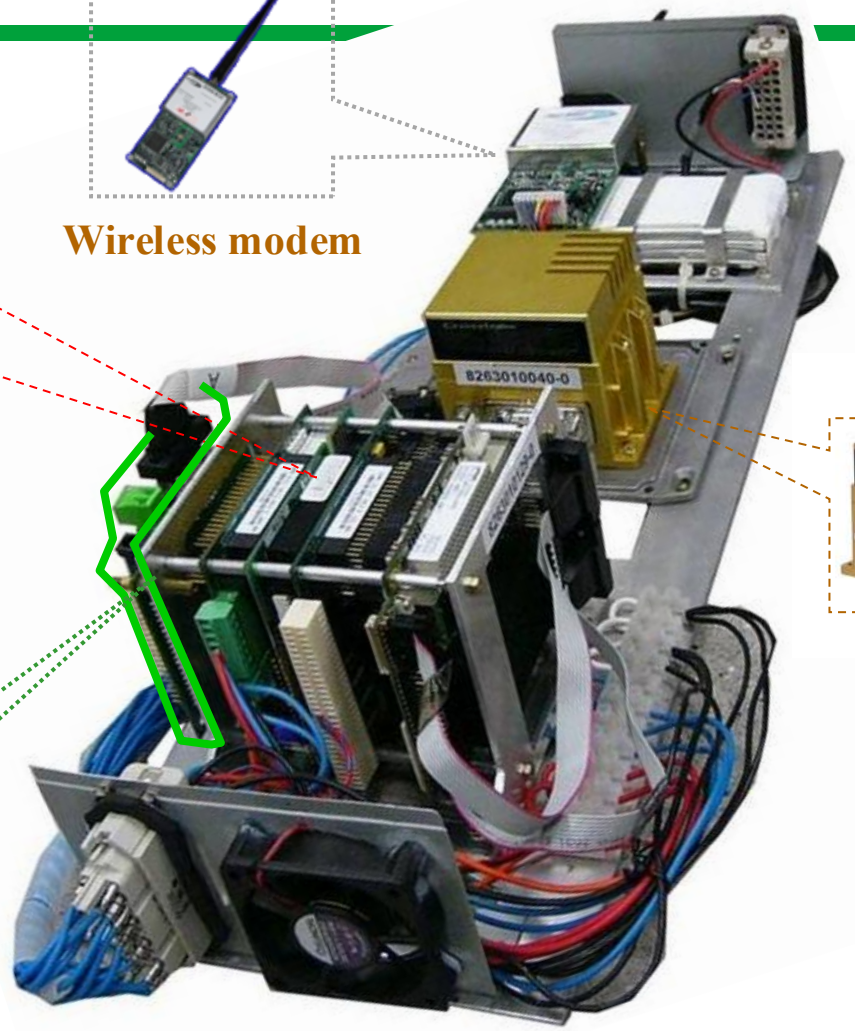
1. Main processing board
2. Serial communication board
3. Data acquisition board
4. DC-to-DC converter board



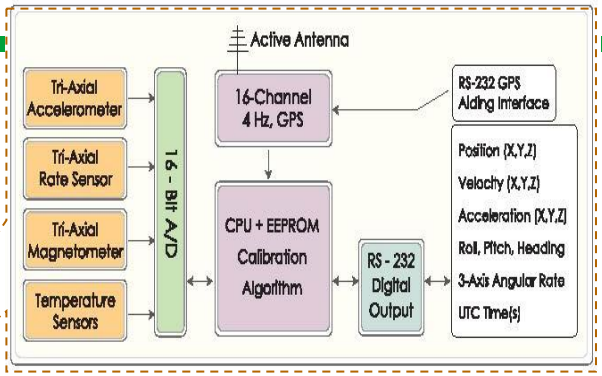
Servo controller



Wireless modem



Avionics System

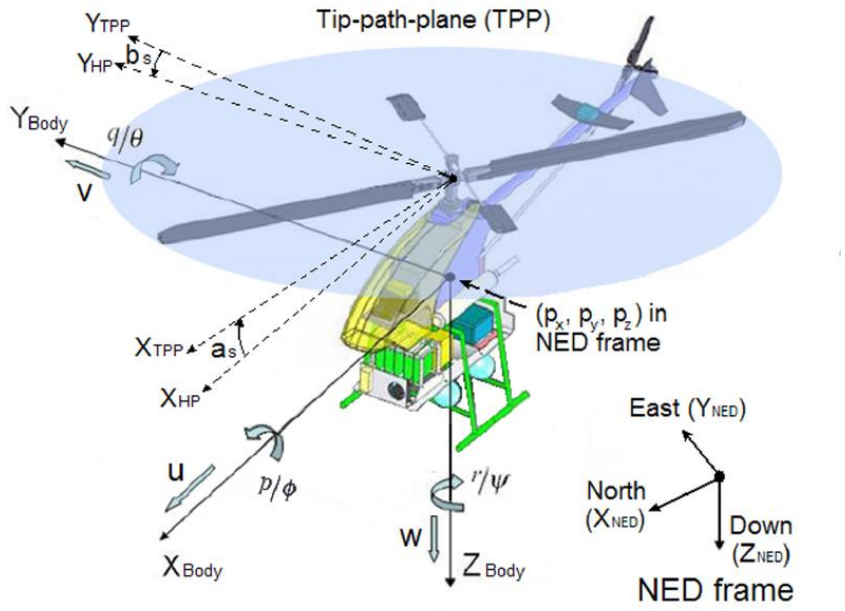




Inertial Measurement Unit (IMU):

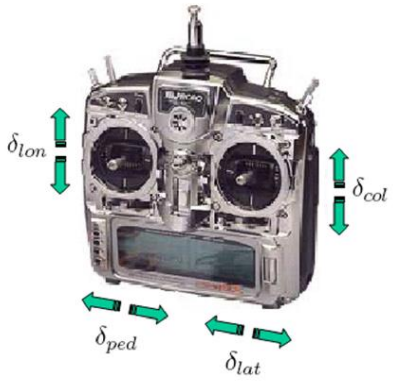
1. Accelerations & velocities
2. Rotating angles
3. Heading angles
4. 3D-GPS signals

Dynamic modelling



Variable	Physical description		Unit
p_x, p_y, p_z	Position vector along NED-frame x, y, and z axes	位置	m
u, v, w	Velocity vector along body-frame x, y, and z axes	速度	m/s
p, q, r	Roll, pitch, and yaw angular rates	角速度	rad/s
ϕ, θ, ψ	Euler angles	欧拉角	rad
a_s, b_s	Longitudinal and lateral tip-path-plane (TPP) flapping angle		rad
$\delta_{ped,int}$	Intermediate state in yaw rate gyro dynamics		NA
δ_{lat}	Normalized aileron servo input $(-1, 1)$	副翼	NA
δ_{lon}	Normalized elevator servo input $(-1, 1)$	升降	NA
δ_{col}	Normalized collective pitch servo input $(-1, 1)$	俯仰	NA
δ_{ped}	Normalized rudder servo input $(-1, 1)$	方向舵	NA



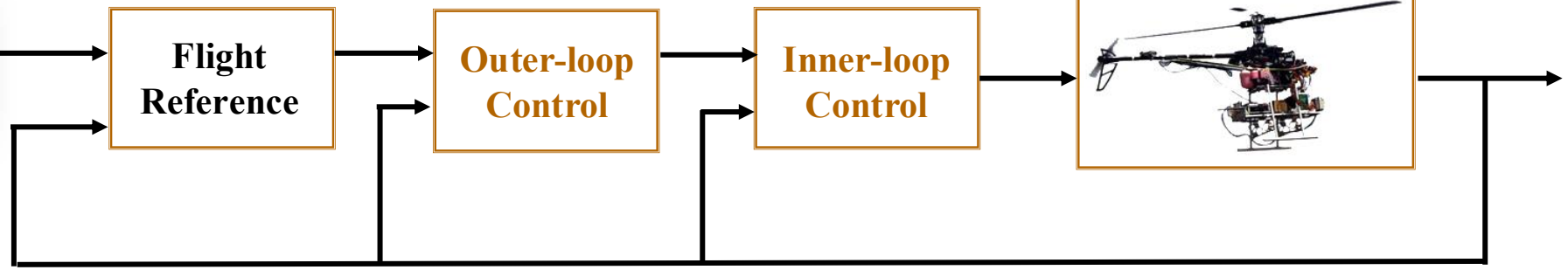


First-principles modeling approach is adopted to obtain an accurate nonlinear model in full envelope, which includes:

- kinematics
- 6 DOF rigid-body dynamics
- main rotor flapping dynamics
- yaw rate gyro dynamics

Flight control system

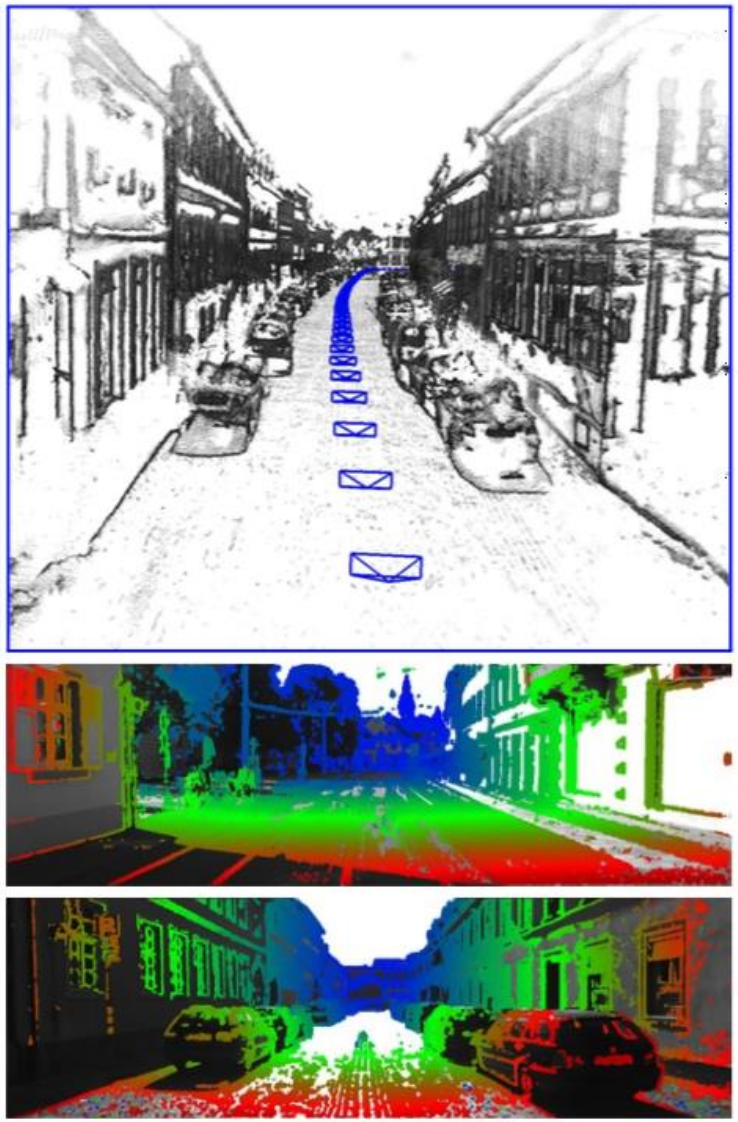


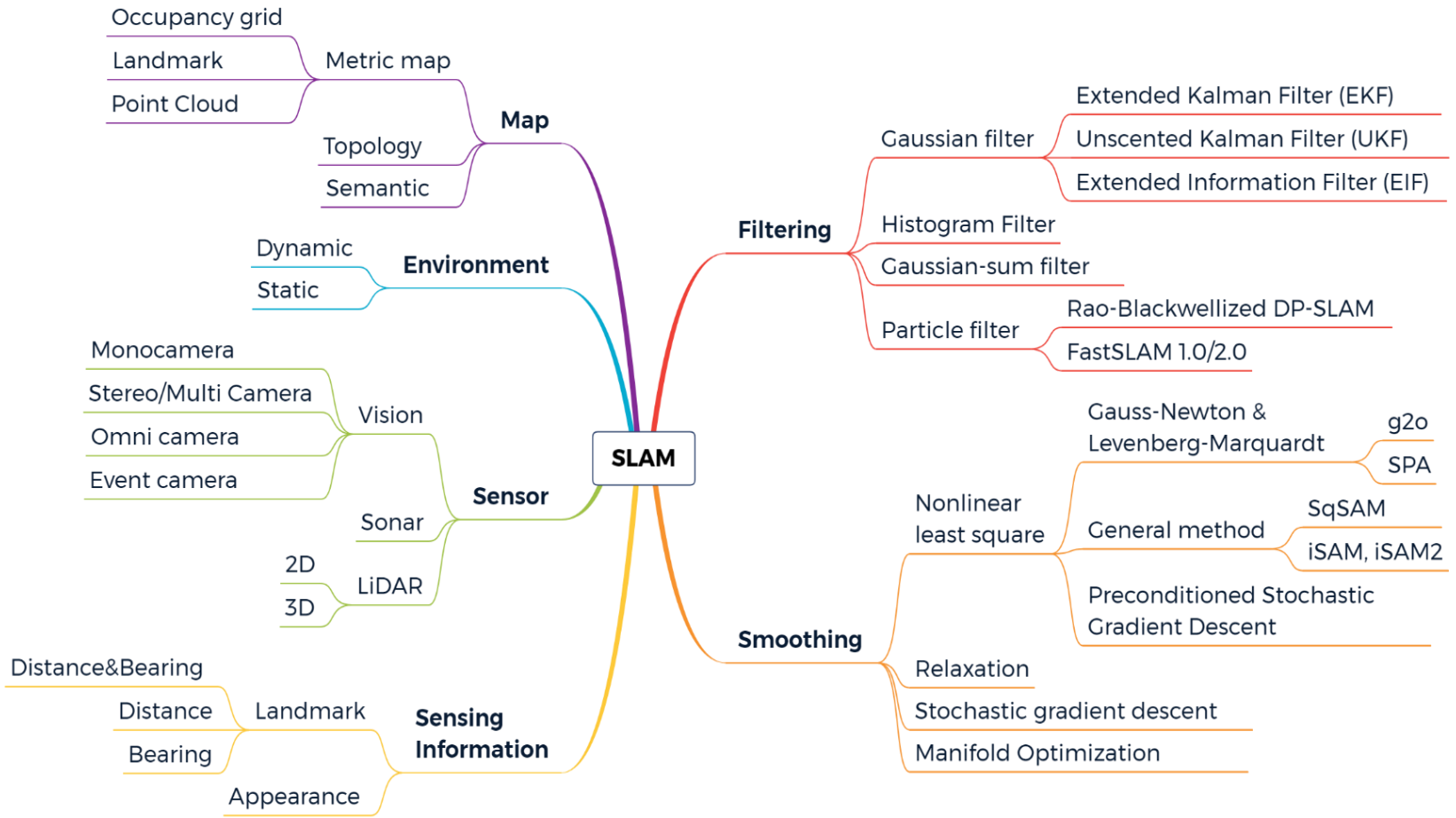


- The inner loop control is to stabilize the aircraft attitude
- The outer loop control is to control the aircraft position and velocity tracking the flight references
- Flight references are generated by flight missions and motion planning

- Inner Loop control
  - PID Control (commonly used)
  - Optimal Control
  - Robust Control
  - Nonlinear Control
- Outer Loop control
  - PID Control (commonly used)
  - Pole placement
  - RPT Control
  - Robust Control

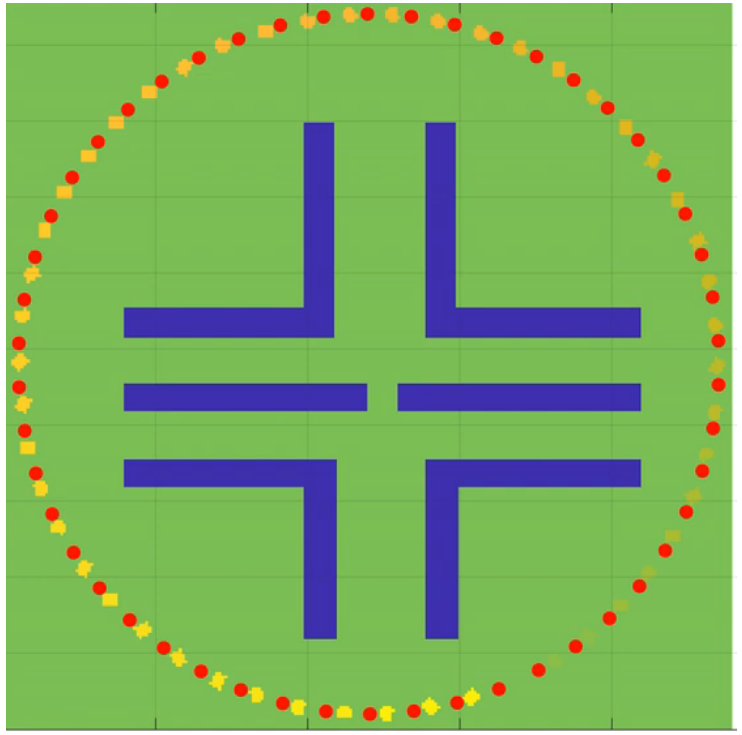
Positioning / Simultaneous localization and mapping (SLAM)



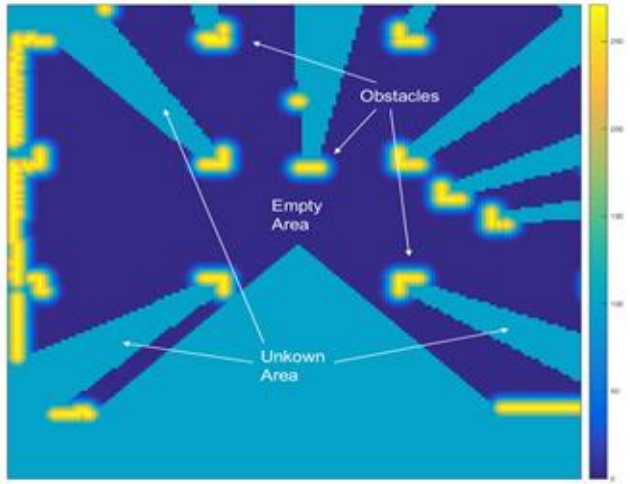


Motion planning

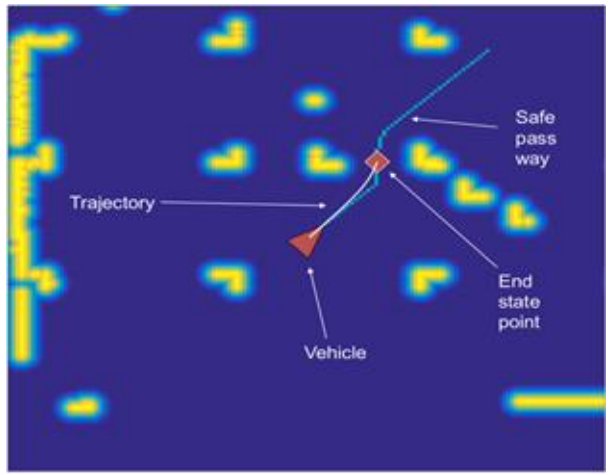
Motion planning is to break down a desired movement task, generated from the **task planning** module, into discrete segments that satisfy movement constraints, such obstacle avoidance in cluttered environments, the limitations in space and those associated with the velocity and acceleration of the vehicle. A motion planning algorithm would produce an appropriate position, velocity, acceleration and jerk reference signals for the **flight control systems.**



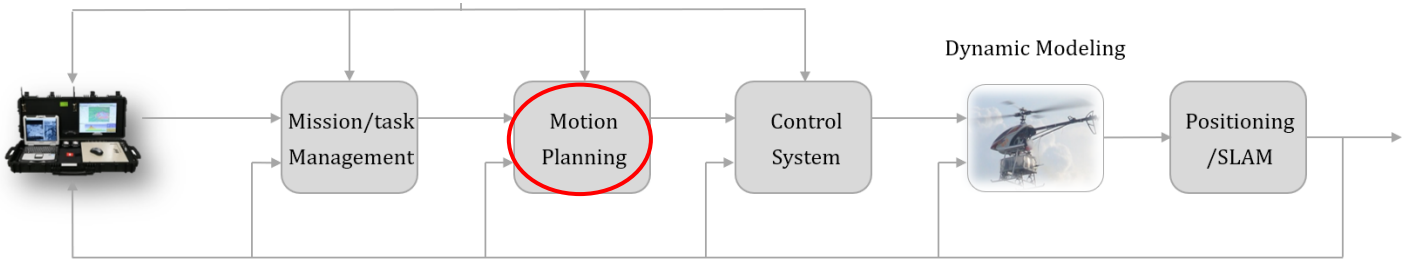
An illustrative example



Path planning



Trajectory generation

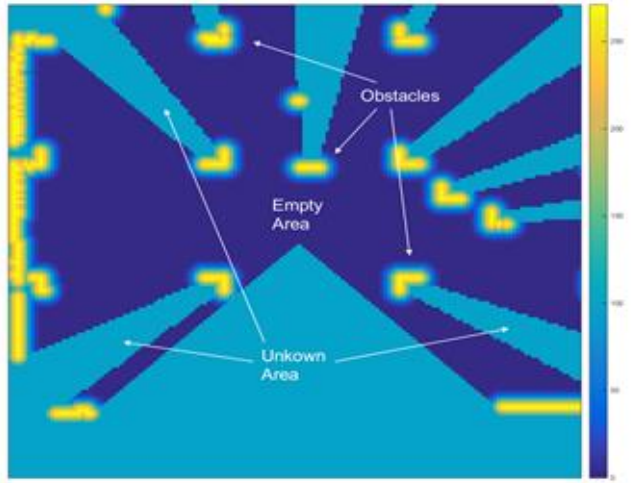


Layered approach for motion planning

分层运动规划方法

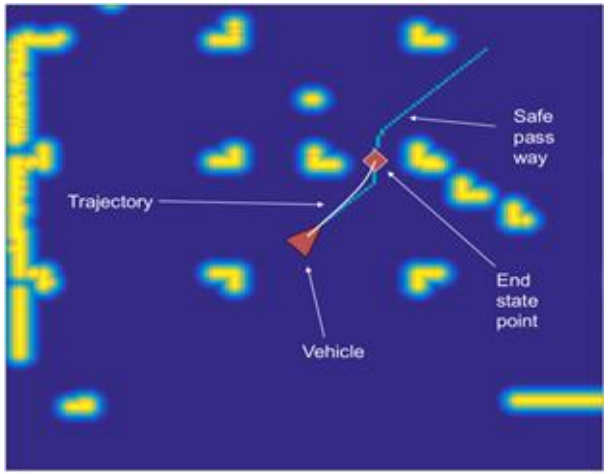
全局规划

- Topological planner (Global planner)
  - A* algorithm.
  - D* lite algorithm.
  - JPS (Jump point search, an improved version of A*)
  - FMT (Fast marching tree)
  - RRT* (Rapid random tree*)
  - BIT* (Batch informed tree*)



局部规划

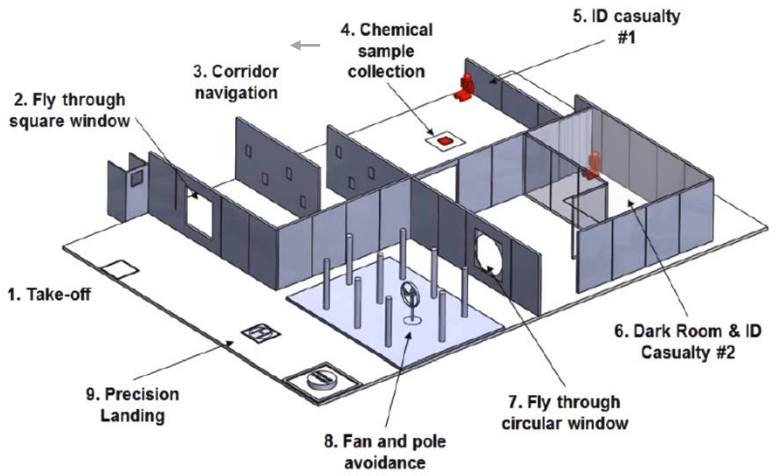
- Trajectory planner (Local planner)
  - Direct optimization-based methods
  - Polynomial spline methods
  - Optimization in parameterized space
  - B-spline approximation methods
  - Jerk limited trajectory optimization-based methods

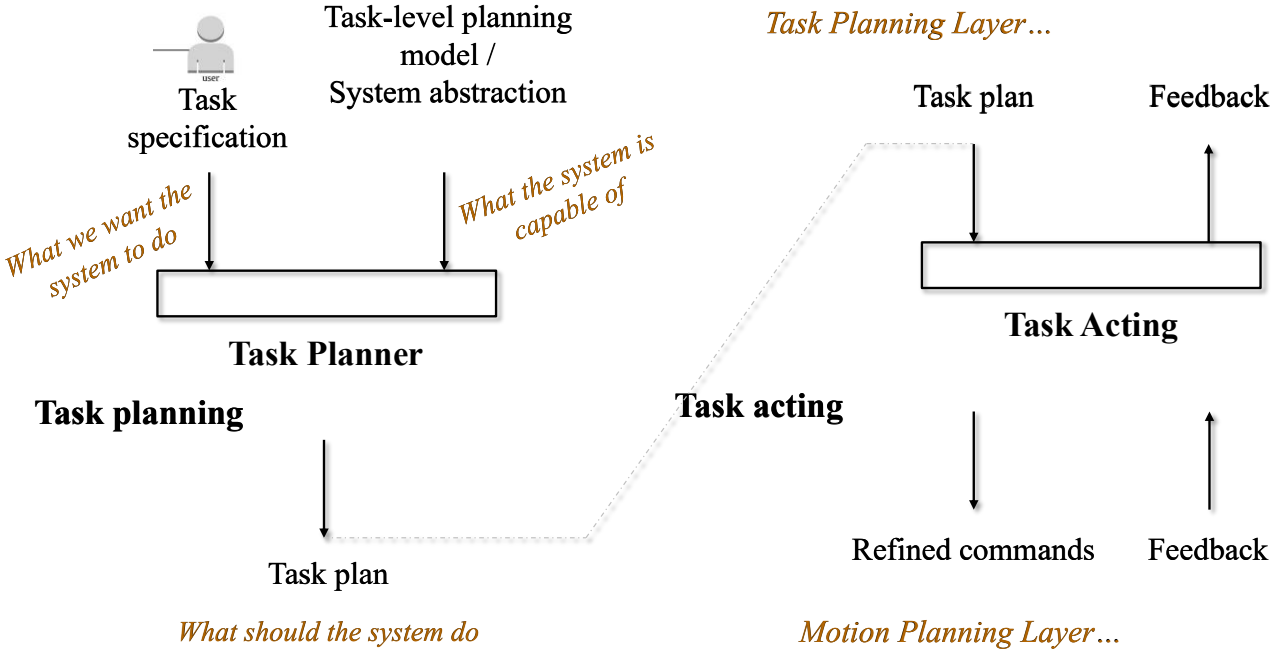


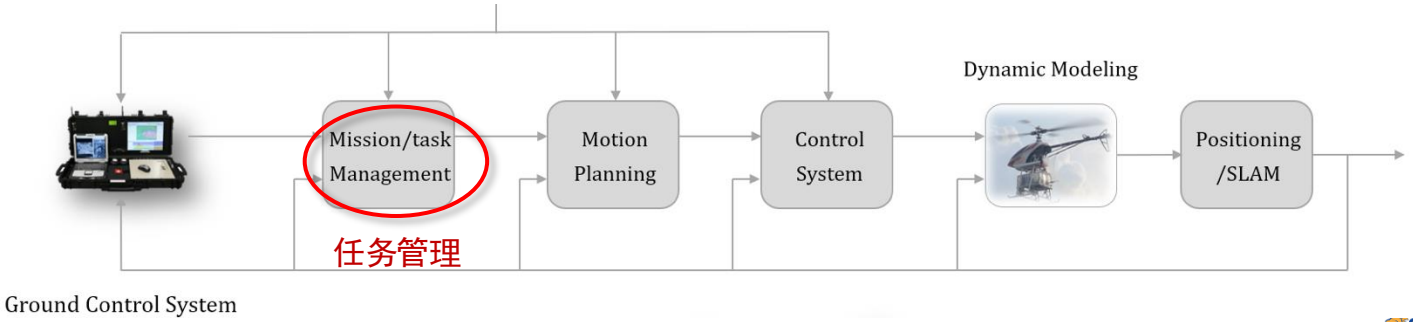
Task planning

Task planning is to translate a mission into operations that may allow the successful execution of the mission, involving the planning of sensory operations, gross-motion planning with collision avoidance, and fine-motion planning with specific tasks.

Example: Indoor Search & Rescue



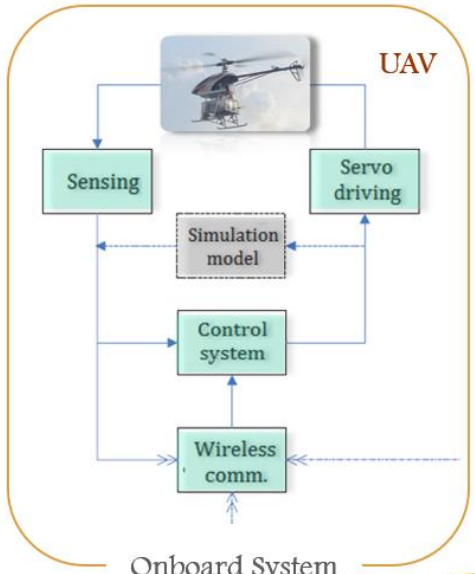




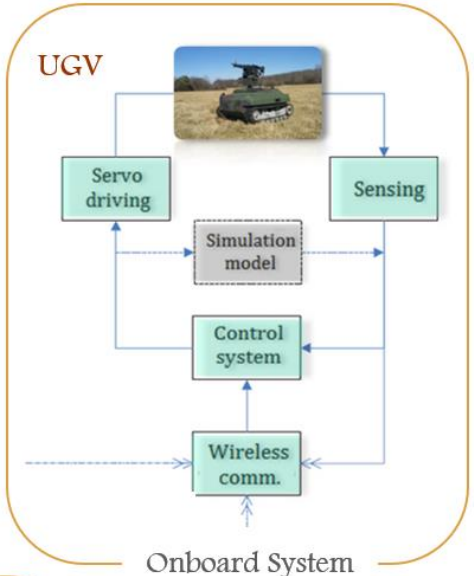
Hidden layer of unmanned systems





















Challenges: Communications and energy supply!

Multi-agent systems



Formation



Task coordination

Homogeneous Systems



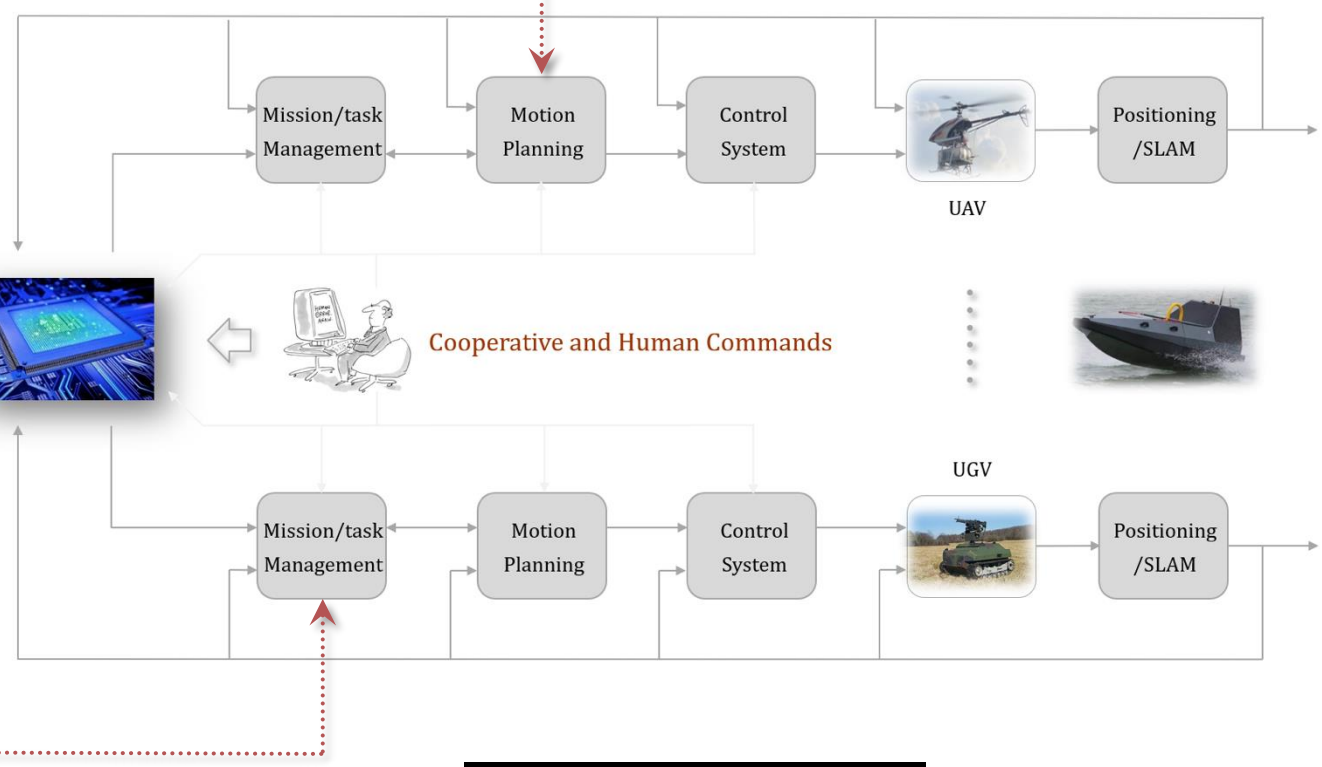


vs

Cooperative and Human Commands



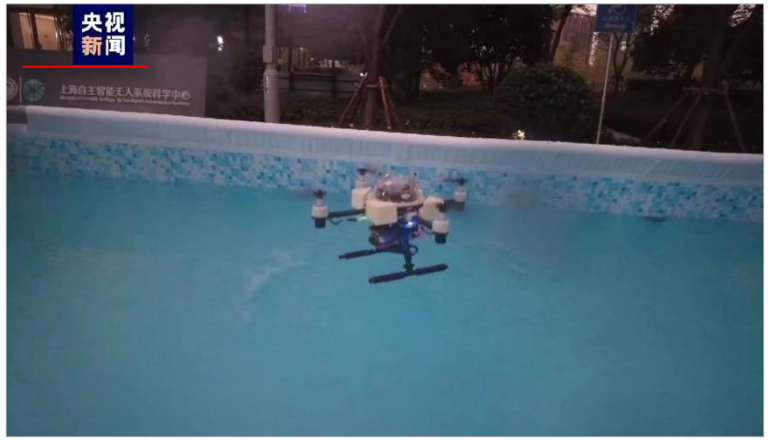
Heterogeneous Systems



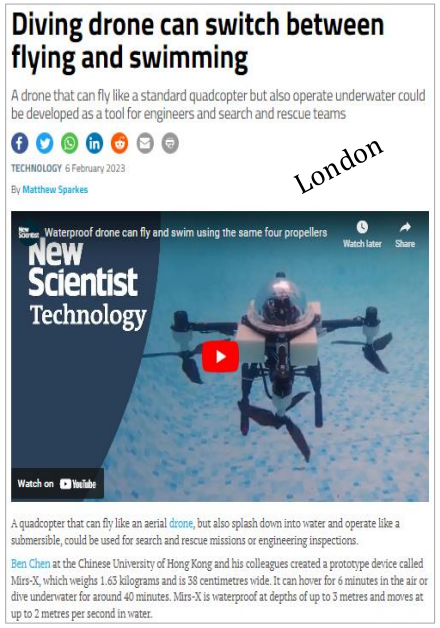
Multi-agent systems?

Some actual unmanned systems









Diving drone can switch between flying and swimming

A drone that can fly like a standard quadcopter but also operate underwater could be developed as a tool for engineers and search and rescue teams

TECHNOLOGY 6 February 2023

By Matthew Sparkes

London

Waterproof drone can fly and swim using the same four propellers

New Scientist Technology

Watch on YouTube

A quadcopter that can fly like an aerial drone, but also splash down into water and operate like a submersible, could be used for search and rescue missions or engineering inspections.

Ben Chen at the Chinese University of Hong Kong and his colleagues created a prototype device called Mirs-X, which weighs 1.63 kilograms and is 38 centimetres wide. It can hover for 6 minutes in the air or dive underwater for around 40 minutes. Mirs-X is waterproof at depths of up to 3 metres and moves at up to 2 metres per second in water.



NEW ATLAS

DRONES

TJ-FlyingFish drone flies through the air and "swims" underwater

By Ben Coxworth

February 07, 2023

North America

VIEW 2 IMAGES



"同济飞鱼"：无人机+潜水器的创新组合

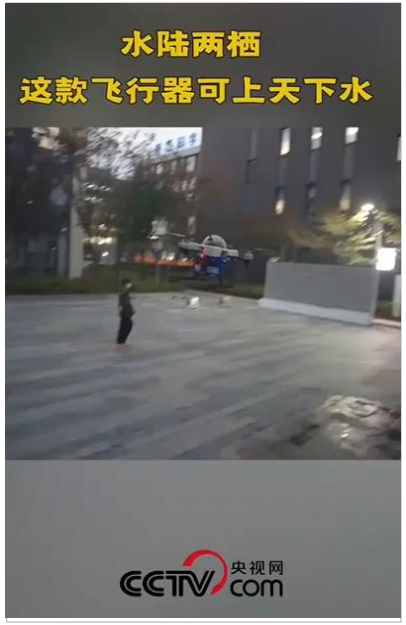
无人机 2023-02-11 16:57

搜救新神器？这款无人机能潜水

环球无人机 81

光明日报记者 孟歆迪

能在空中飞行，也能在水中畅游。小小的飞行器可以轻轻松松做到水、空无缝切换。近日，一项名为"TJ-FlyingFish"（同济飞鱼）的水空跨域巡航器原型机，先后被《新科学家》、北美New Atlas等





2023年5月13日 星期六 大公報

中大與上海同濟研水空兩棲巡航器

【大公報訊】記者廖敏烟報道：近期，香港中文大學與上海同濟大學共同研發出一款水空兩棲巡航器原型機，名為「同濟飛魚」。這款巡航器原型機於今年二月發布，可以在空中飛行，亦可在水下暢遊探索。中大機械及自動化工程學系陳本美教授表示，預計「同濟飛魚」推向市場不需太長時間。如有機會，希望明年可以正式推出。

據介紹，「同濟飛魚」具備一個中央的圓頂形機身和四個附有電機或螺旋槳模組的支臂，重量為1.63公斤。其擁有跨領域定位和導航系統，包括全球定位系統、慣性測量單元、深度計和速度測數日誌，可全自動操作整個水陸兩棲的航程，毋須人手操作。

在空中「飛」時，這款無人機的四個支臂會朝上以高速度旋轉。而當它降落在水面成為「魚」，支臂就會朝下，皆以較慢的速度旋轉，幫助推動身拉入水中。在充滿電的情況下，「同濟飛魚」可在空中飛行6分鐘，在水中巡遊40分鐘，最深去到水底3米，最快可在水底速到每秒2米。

可用於勘測與搜救

陳教授提到，在研發「同濟飛魚」時，其中一個具備挑戰性的工作是訪水，而另一個正在努力攻破的難題則是水底導航定位（GPS）。為了更好地在水中進行勘測，該團隊未來將會根據研究需求，在無人機上加裝鏡頭、聲吶探測設備等部件。

而除了挑戰，還有突破。與一般無人機不同的是，「同濟飛魚」機身每個支臂末端均配上一個設有變速箱的推進裝置，可以用兩種速度獨立旋轉。該裝置可因應水空介質進行切換，達到最佳運行效果。

陳教授總指，「同濟飛魚」是團隊研究工作的重要里程碑。「我們期待它在未來被廣泛應用。」目前，這款無人機可運用於航空和水中勘測、遙距感應和災難搜救工作等場景。

▲陳本美教授（左圖後排中）表示，預計「同濟飛魚」最快可在明年推出市場。 大公報記者廖敏烟攝



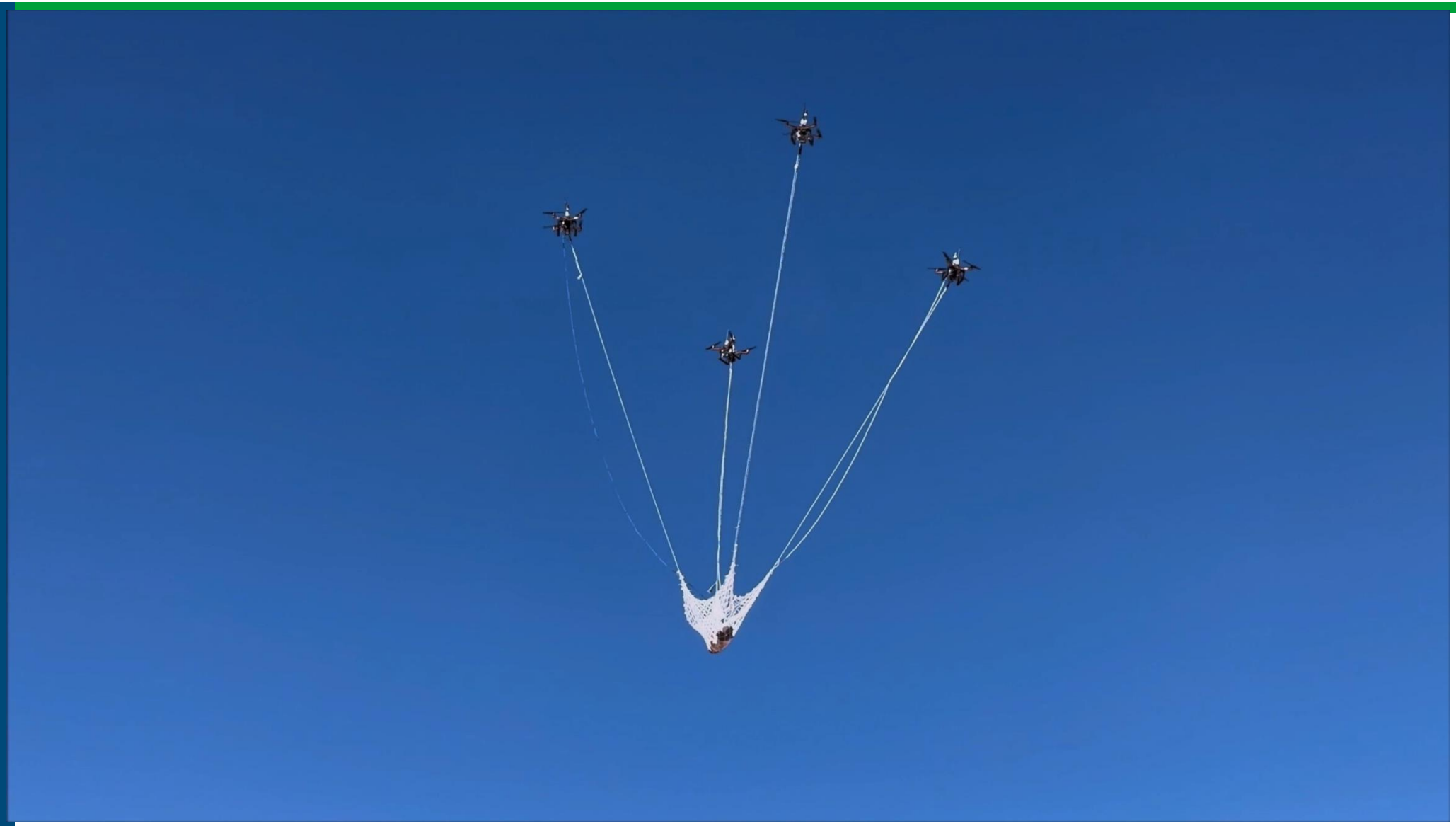
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中国式现代化是中国共产党领导的社会主义现代化

Some actual unmanned systems



Key techniques involved: cooperative control

Some actual unmanned systems



Key techniques involved: perception and planning!



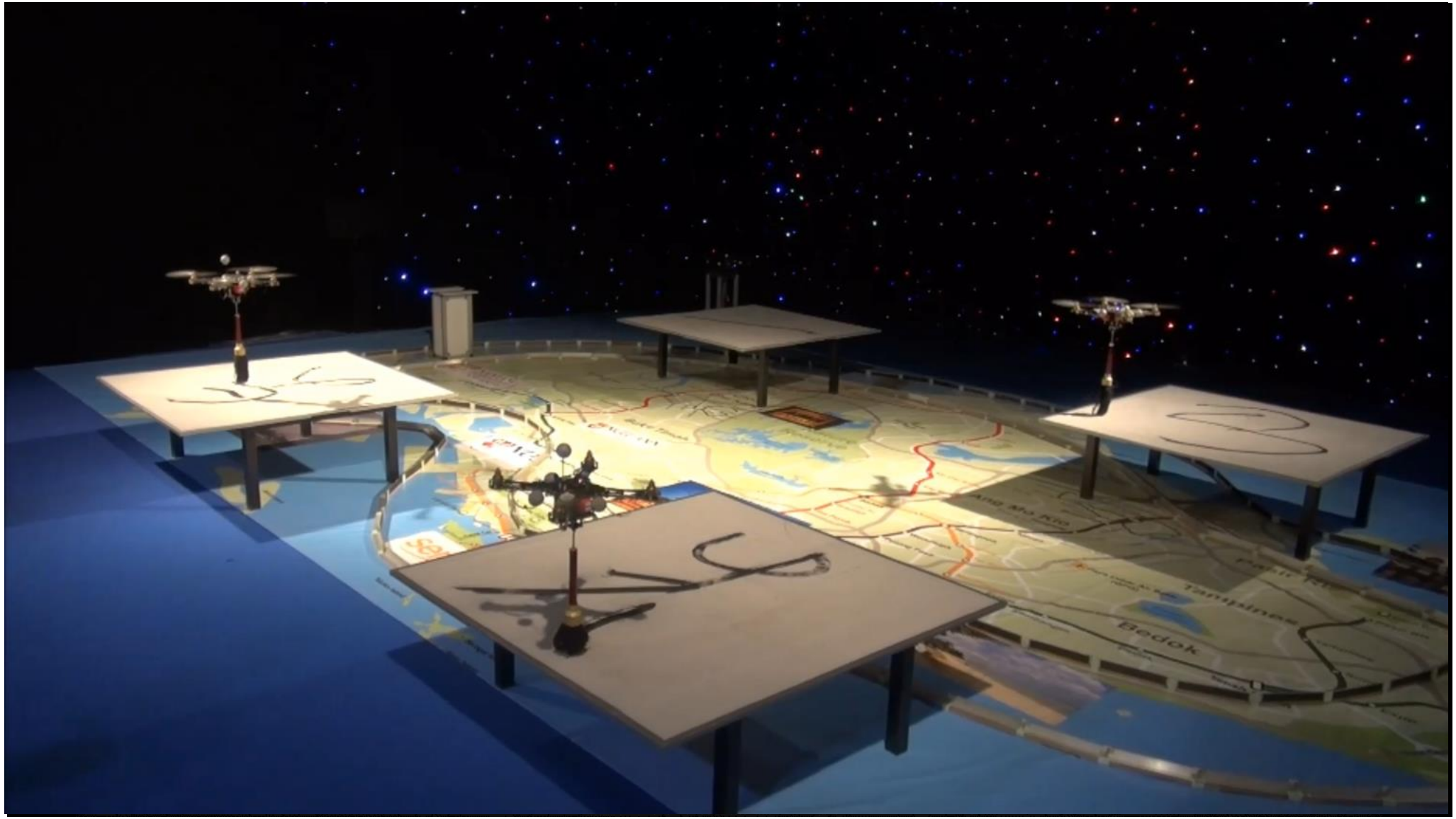
Some actual unmanned systems



Key techniques involved: perception, task and motion planning!



Some actual unmanned systems



Key techniques involved: motion planning!

Some actual unmanned systems



Key techniques involved: localization and motion planning!

Some actual unmanned systems





Key techniques involved: perception, task and motion planning!

Some actual unmanned systems

Courtesy of MicroHiggs Technologies



Key techniques involved: hardware platform design!

Some actual unmanned systems

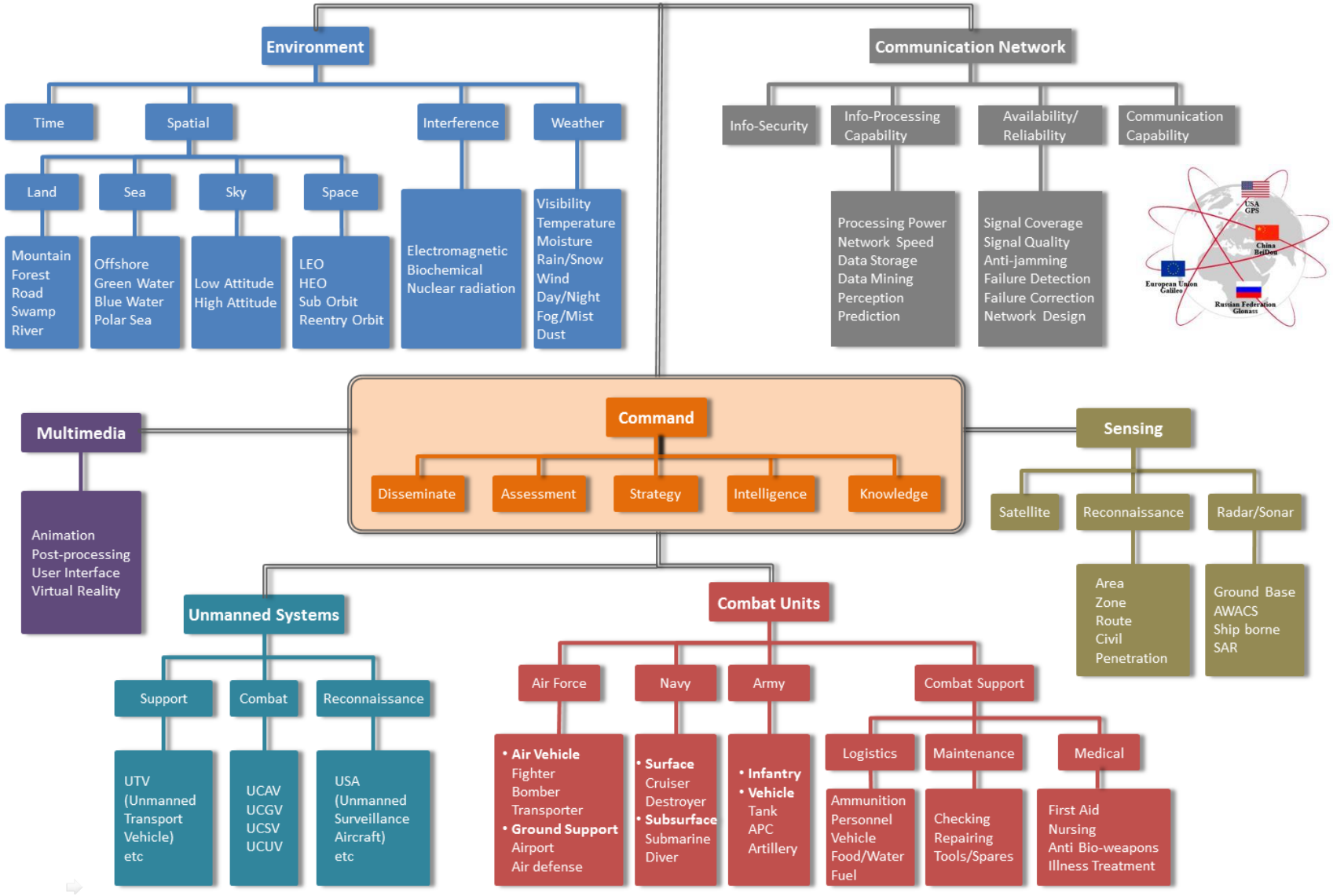




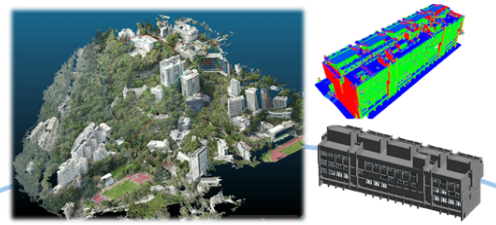




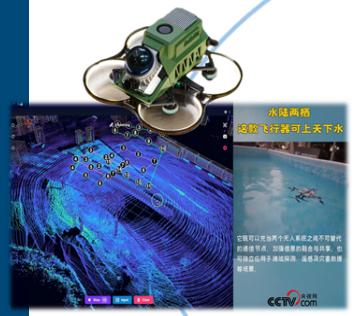
Future of unmanned systems…



Future of unmanned systems…



3D Reconstruction and Information Modeling



Embodied AI-assisted
Unmanned Systems

Smart City Applications





LLM-assisted
Digital Twin System

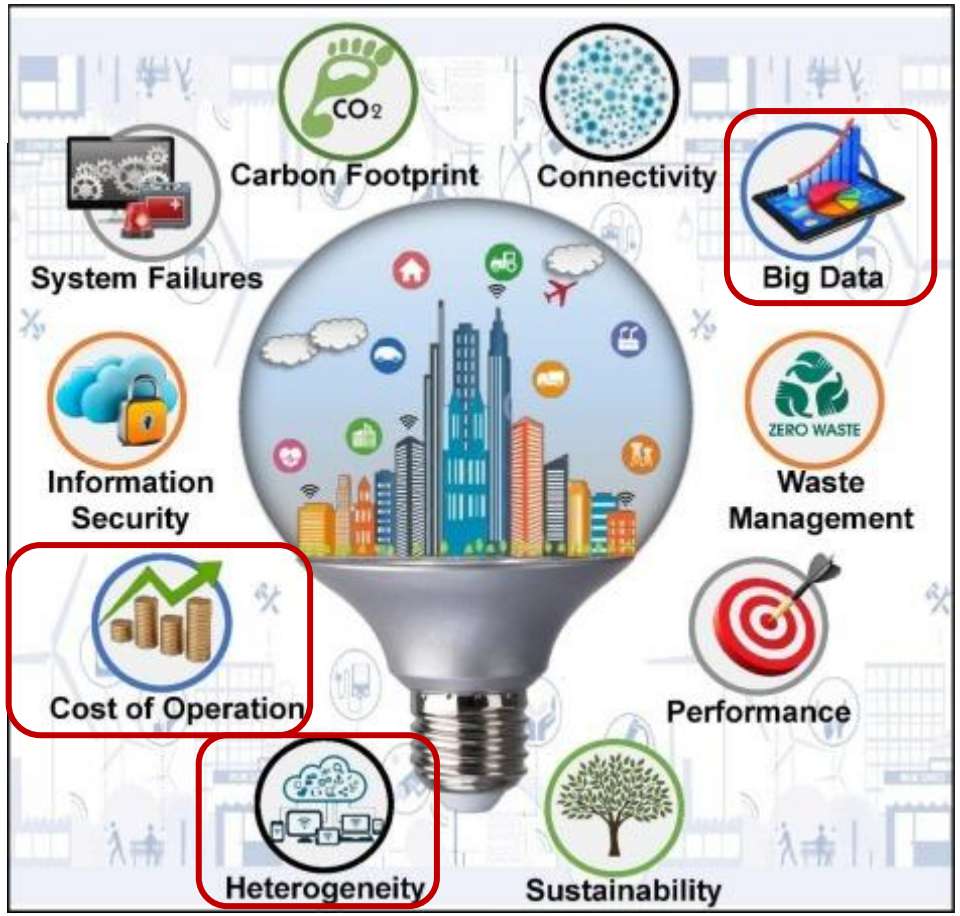
Smart city to Smart Ocean







Challenges for smart city



Silva BN, Khan M, Han K. Towards sustainable smart cities: A review of trends, architectures, components, and open challenges in smart cities. Sustainable Cities and Society. 2018;38:697-713


THANK
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